

1. Student Query

Student asks a question / submits a problem

2. Context Understanding

Retrieve relevant context from conversation history and resources

3. Generate Socratic Question / Hint

LLM generates a guided question or hint instead of giving the direct answer

4. Student Response

Student attempts to answer or responds to the guidance

5. Response Evaluation

Evaluate the response and assess conceptual understanding

6. Concept Understood?

No

7A. Provide Further Guidance

Provide more hints, simpler questions, or alternative explanations

Yes

7B. Generate Formative Feedback

Provide constructive feedback and reinforce correct understanding

8. Update Learning Progress

Store interaction, update learning progress and session context

